ROS2 개발 환경 구축

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1 영상으로.(영상은 Humble 버전임에 유의!)

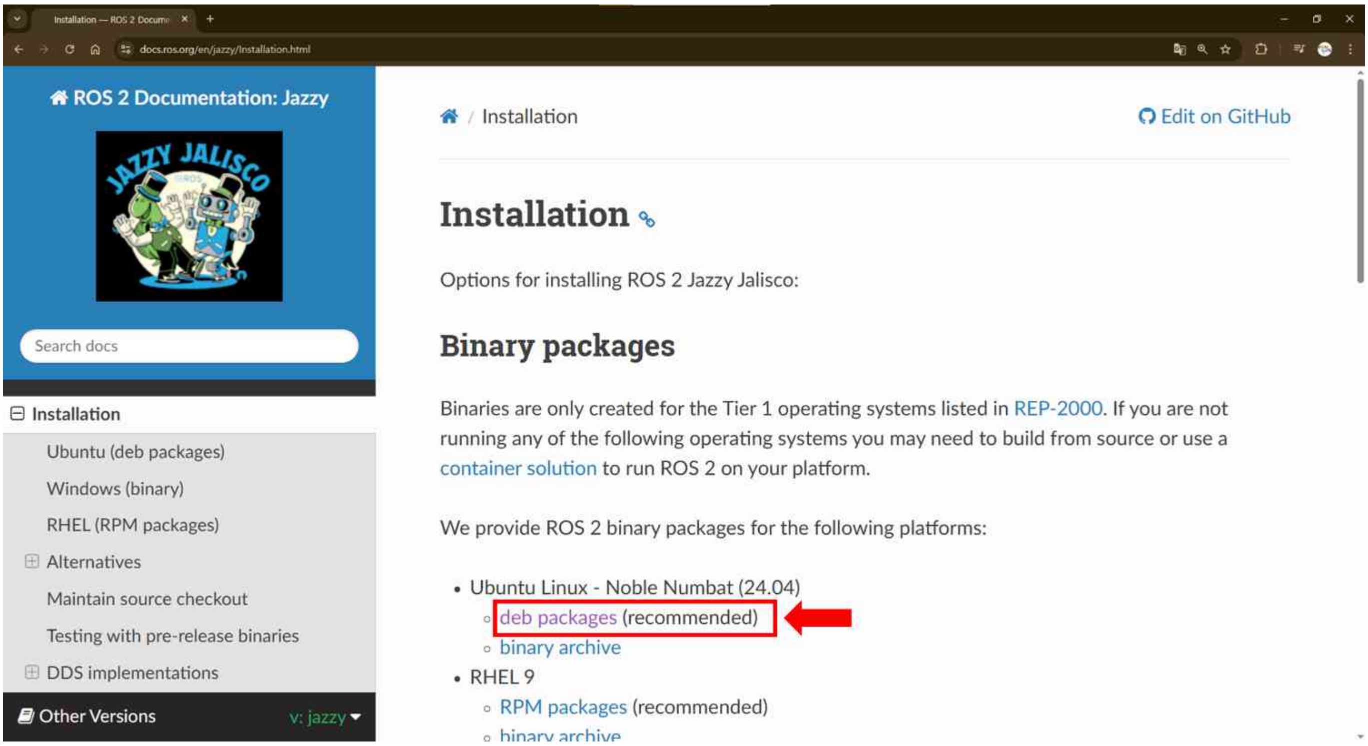
ROS2 개발 환경 구축

우리는 우분투 24.04에서 시작합니다.

ROS2 Jazzy 설치 페이지

https://docs.ros.org/en/jazzy/Installation.html (https://docs.ros.org/en/jazzy/Installation.html)

Debian Packages (Ubuntu) 선택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docs.ros.org/en/jazzy/Installation.html
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JAZZY JALISCO
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Installation
Options for installing ROS 2 Jazzy Jalisco:
Binary packages
Binaries are only created for the Tier 1 operating systems listed in REP-2000. If you are not running any of the following operating systems you may need to build from source or use a container solution to run ROS 2 on your platform.
We provide ROS 2 binary packages for the following platforms:
Ubuntu Linux - Noble Numbat (24.04)
deb packages (recommended)
binary archive
RHEL 9
RPM packages (recommended)
binary archive

설치 과정 따라가기

Ubuntu (deb packages) — RO

docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html

ROS 2 Documentation: Jazzy

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Other Versions v: jazzy

/ Installation / Ubuntu (deb packages) — Edit on GitHub

Ubuntu (deb packages)

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Deb packages for ROS 2 Jazzy Jalisco are currently available for Ubuntu Noble (24.04). The target platforms are defined in REP 2000.

Resources

- Status Page:
  - ROS 2 Jazzy (Ubuntu Noble 24.04): amd64, arm64
- Jenkins Instance
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System setup

Set locale

Make sure you have a locale which supports `UTF-8`. If you are in a minimal environment (such as a docker container), the locale may be something minimal like `POSIX`. We test with the following settings. However, it should be fine if you're using a different UTF-8 supported locale.

Set locale

System setup

Set locale

Make sure you have a locale which supports `UTF-8`. If you are in a minimal environment (such as a docker container), the locale may be something minimal like `POSIX`. We test with the following settings. However, it should be fine if you're using a different UTF-8 supported locale.

```
locale  # check for UTF-8

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale  # verify settings
```

locale 설정 확인

```
ham@ham-2404: ~
ham@ham-2404: ~ 80x24
ham@ham-2404:~$ locale
LANG=en_US.UTF-8
LANGUAGE=
LC_CTYPE="en_US.UTF-8"
LC_NUMERIC="en_US.UTF-8"
LC_TIME="en_US.UTF-8"
LC_COLLATE="en_US.UTF-8"
LC_MONETARY="en_US.UTF-8"
LC_MESSAGES="en_US.UTF-8"
LC_PAPER="en_US.UTF-8"
LC_NAME="en_US.UTF-8"
LC_ADDRESS="en_US.UTF-8"
LC_TELEPHONE="en_US.UTF-8"
LC_MEASUREMENT="en_US.UTF-8"
LC_IDENTIFICATION="en_US.UTF-8"
LC_ALL=en_US.UTF-8
ham@ham-2404:~$
```

copy해서 터미널에 붙여넣기 (sudo 조심)

```
locale  # check for UTF-8

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale  # verify settings
```

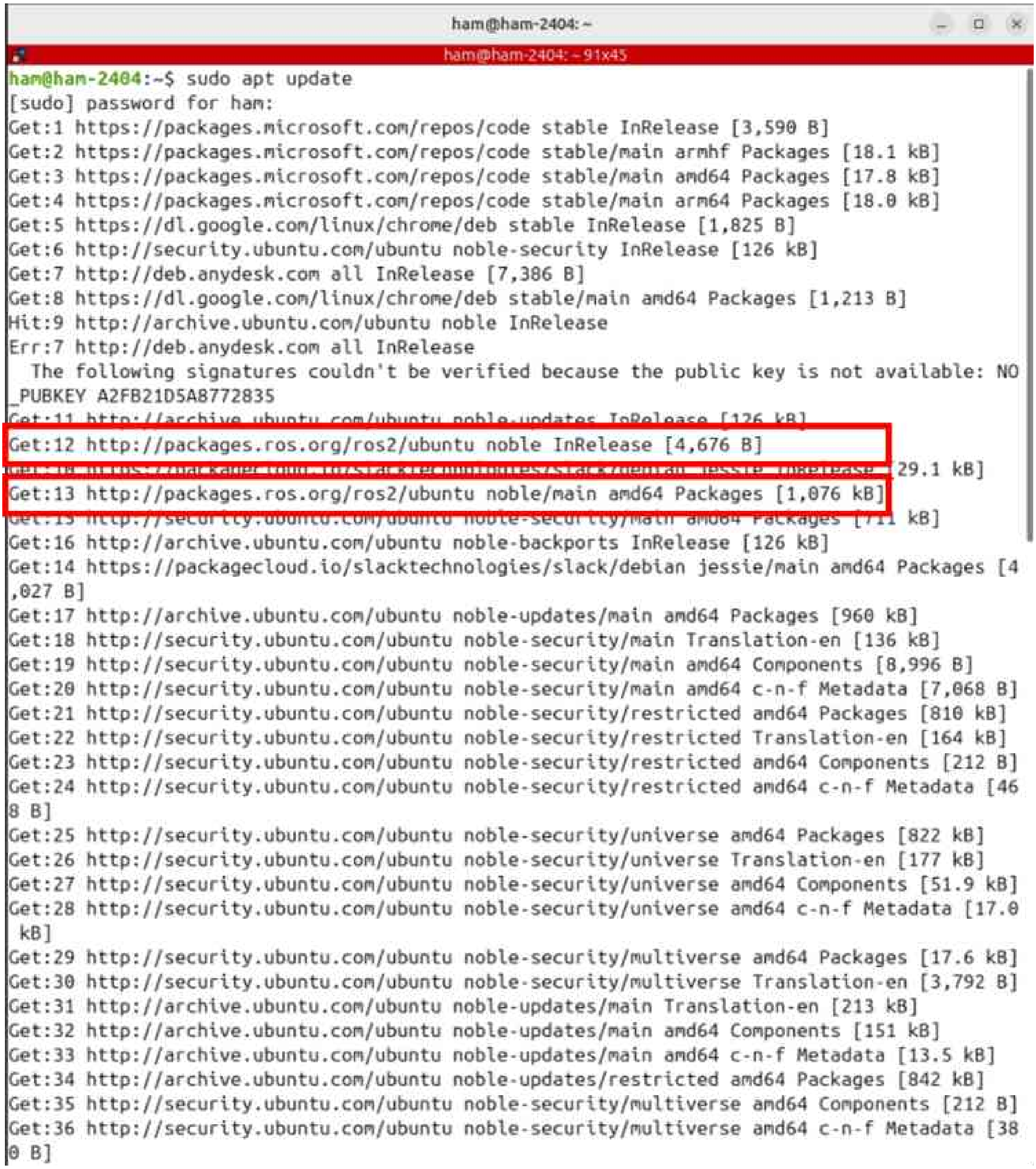
Ubuntu 자체가 제공하는 “universe” 저장소 활성화

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

ROS 2 전용 APT 저장소를 자동으로 등록

```
$ sudo apt update && sudo apt install curl -y
$ export ROS_APT_SOURCE_VERSION=$(curl -s https://api.github.com/repos/ros-infrastructure/ros-apt-source/releases/latest | grep -F "tag_n
$ curl -L -o /tmp/ros2-apt-source.deb "https://github.com/ros-infrastructure/ros-apt-source/releases/download/${ROS_APT_SOURCE_VERSION}/r
$ sudo apt install /tmp/ros2-apt-source.deb
```

apt update 진행



apt upgrade 진행

Code

```
sudo apt upgrade
```

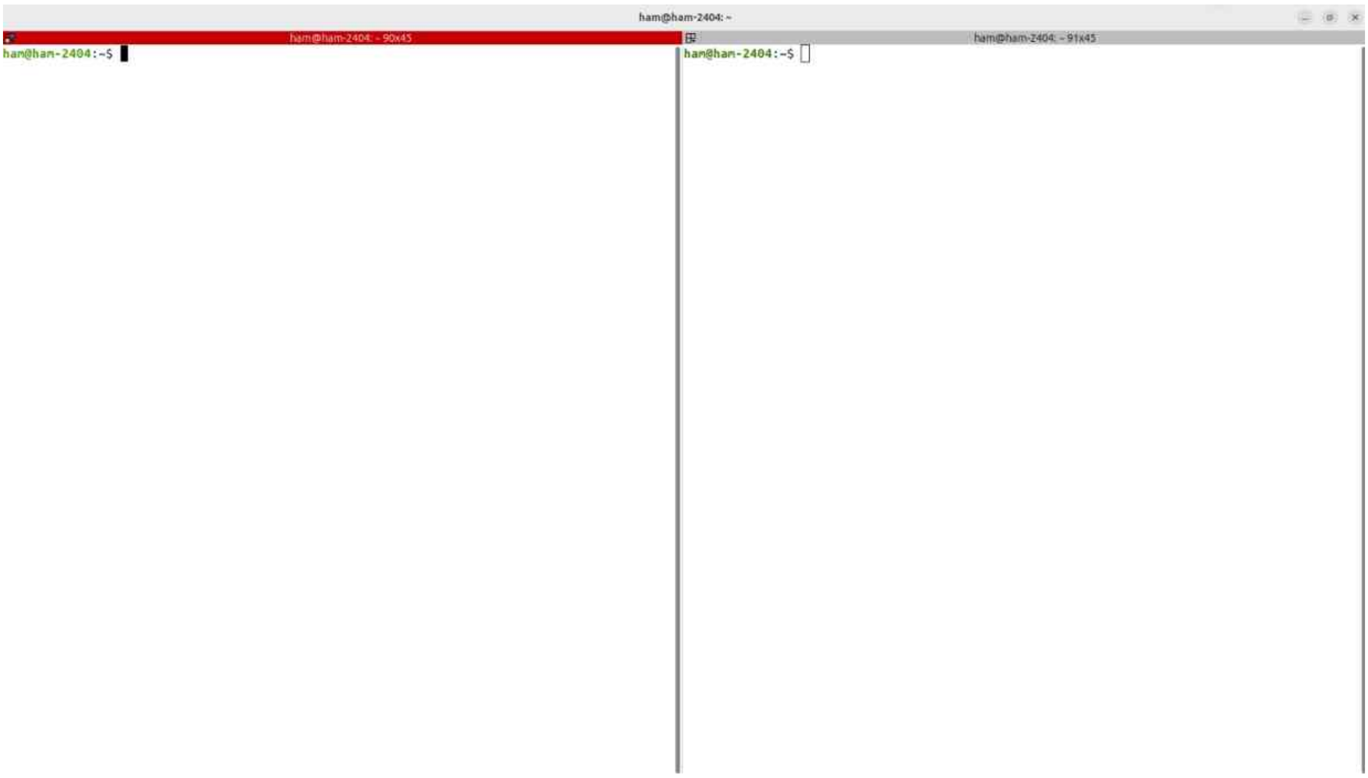
ros-jazzy-desktop 설치 시작

```
sudo apt install ros-jazzy-desktop
```

설치 중

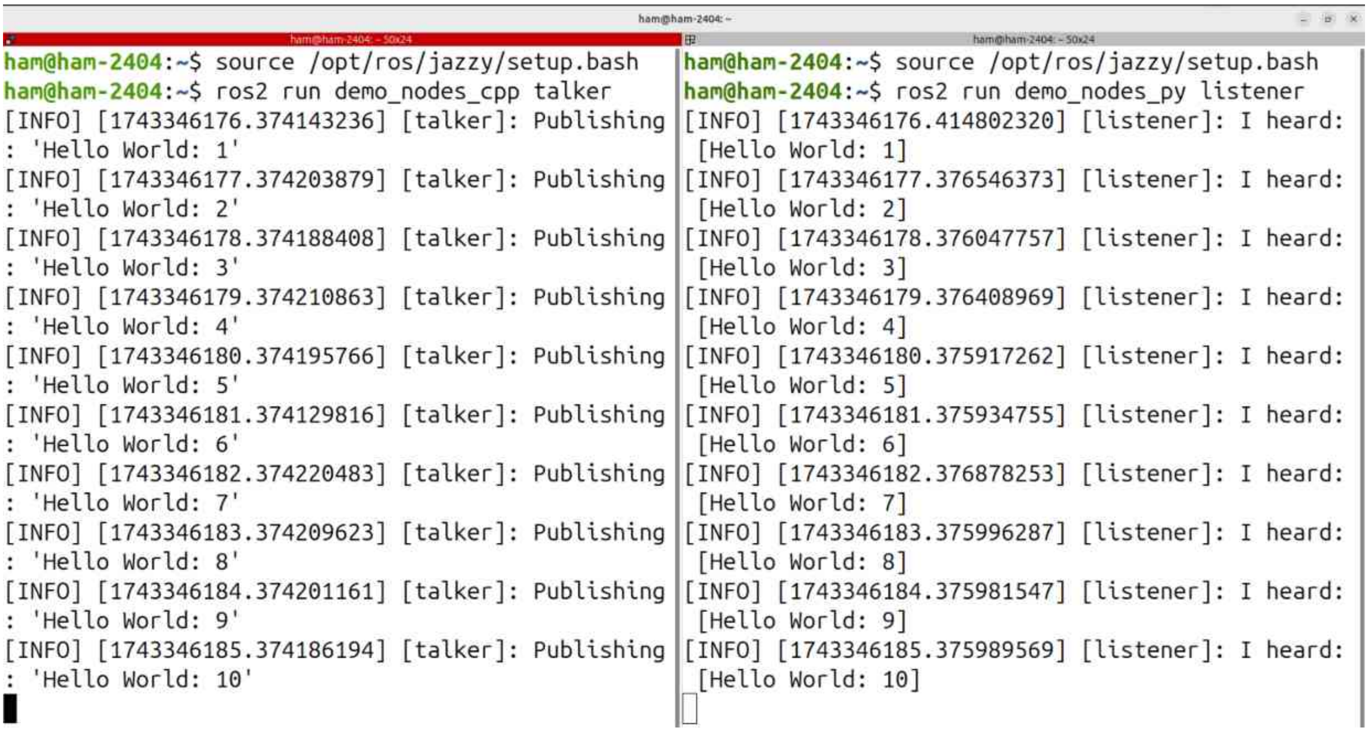
```
  sip-dev tango-icon-theme tcl-dev tcl8.6-dev tk-dev tk8.6-dev
  uncrustify unicode-data unixodbc-common unixodbc-dev uuid-dev
  va-driver-all vdpau-driver-all vtk9 x11proto-dev
  xorg-sgml-doctools xtrans-dev
0 upgraded, 895 newly installed, 0 to remove and 4 not upgraded.
Need to get 516 MB of archives.
After this operation, 2,383 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

설치 후 두 개의 터미널 준비해서



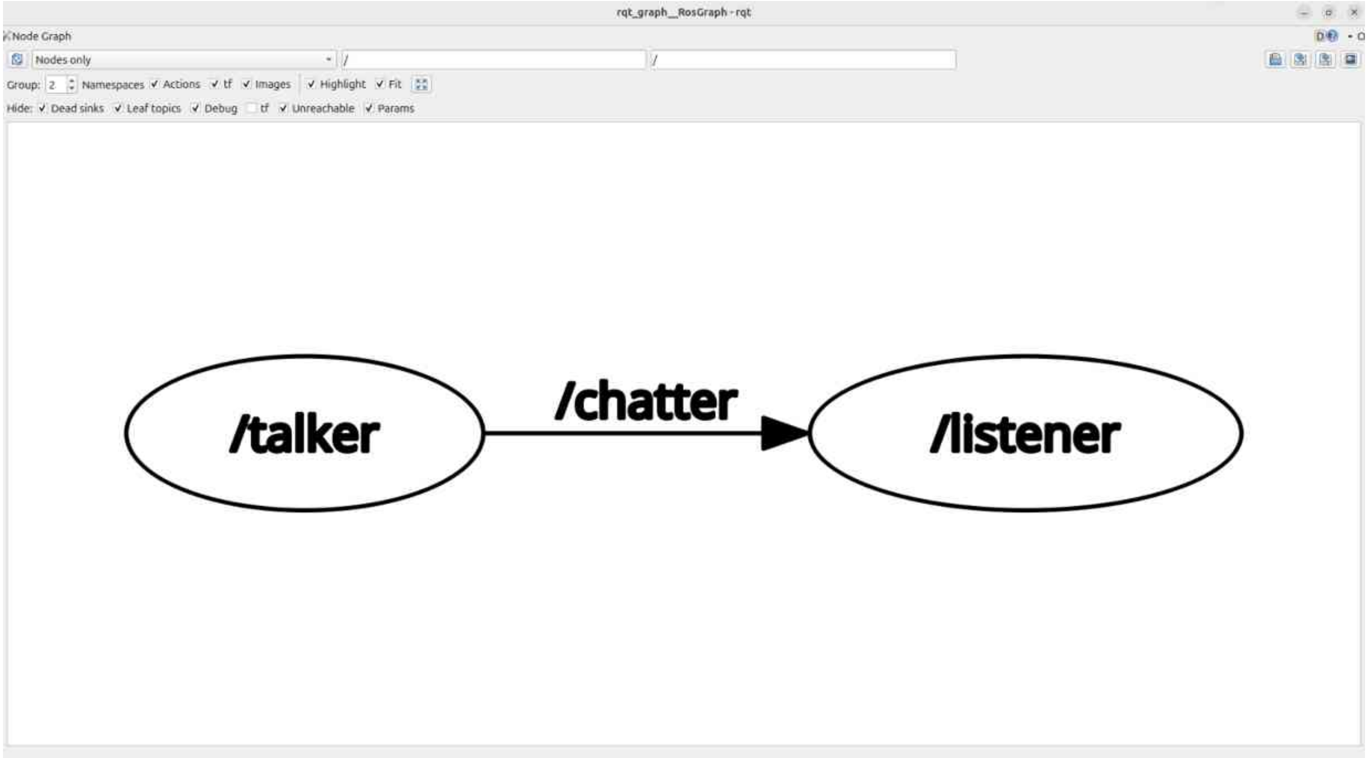
둘다 ROS2 Jazzy의 setup.bash를 source로 부르고

각각 talker와 listener 실행



잘 작동되면 성공적으로 설치된 것 입니다.

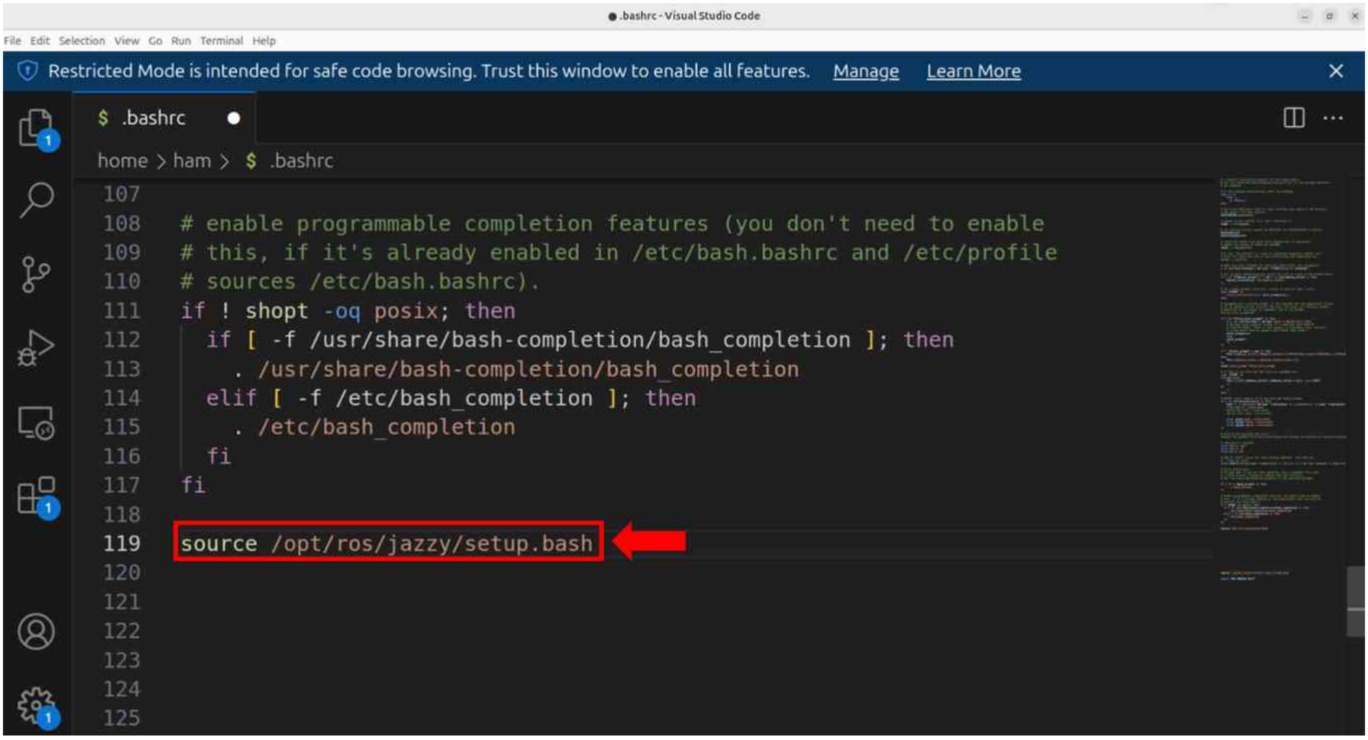
이렇게 연결된 것 입니다.



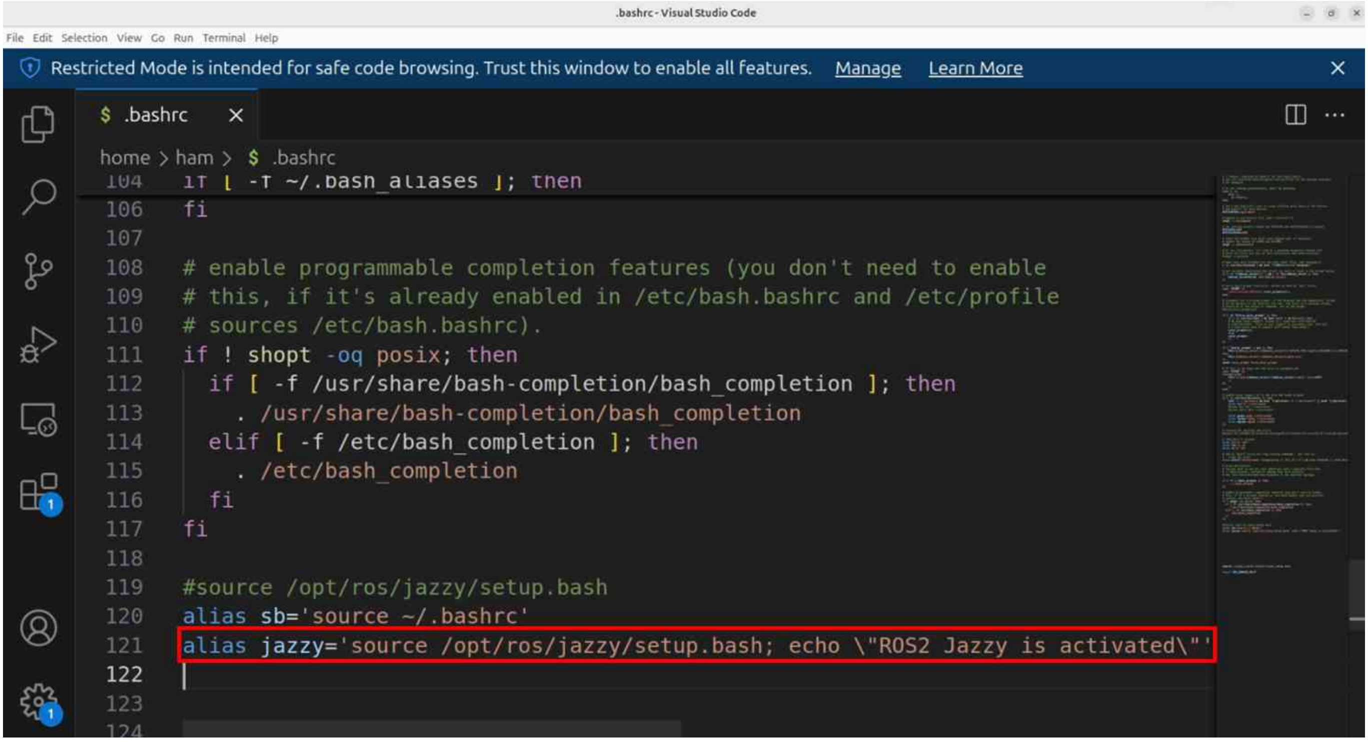
터미널이 실행될 때마다 ROS2 환경을 불러오려면?

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

~/.bashrc 파일의 내용은 터미널이 실행될 때 자동으로 실행



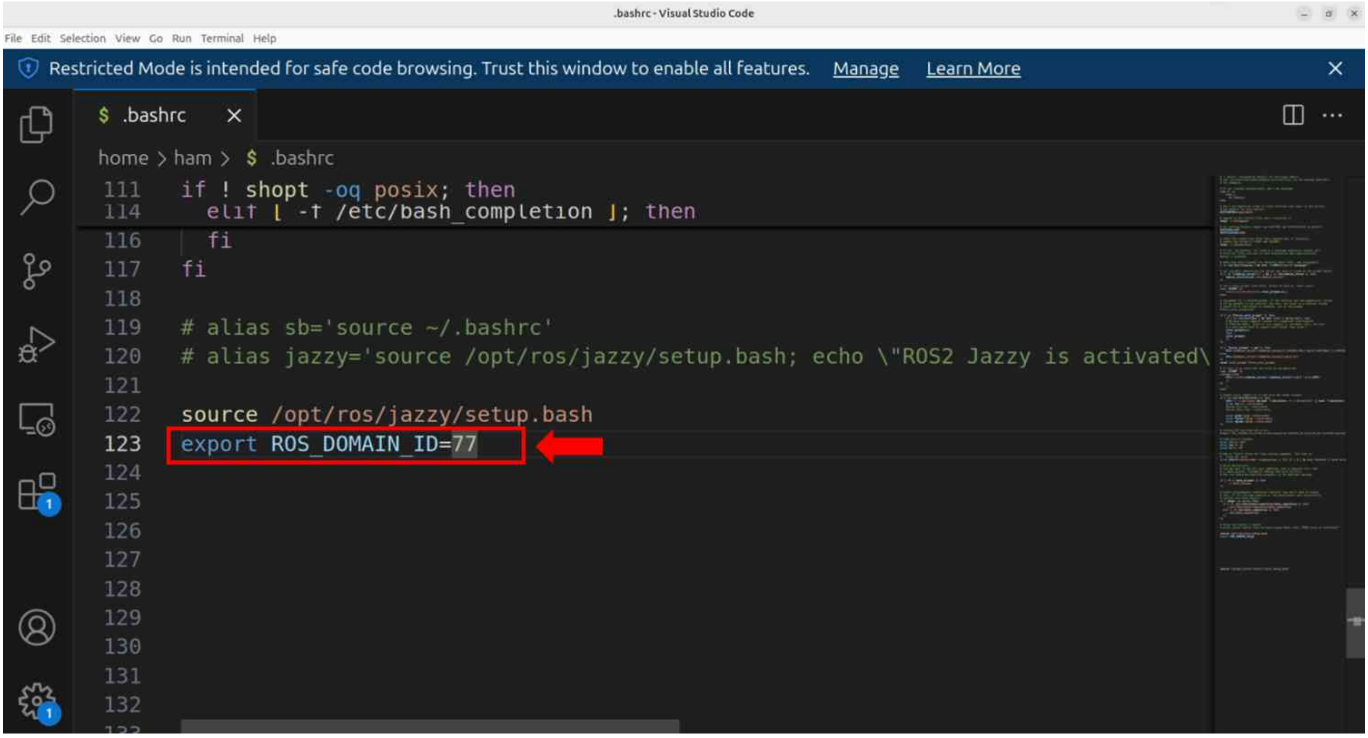
혹은 alias로 필요할 때마다 불러오게 설정



터미널을 새로 열 때마다 등록해둔 단축어를 사용

ROS2_DOMAIN_ID

- 서로의 데이터를 공유하면 문제가 생길 수 있으므로 격리시켜야 한다!



추가 패키지 설치

rosdep 등 개발에 필요한 패키지를 설치

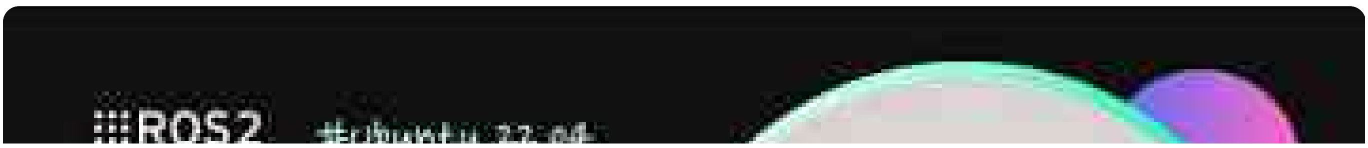
```
sudo apt update && sudo apt install ros-dev-tools
```

Colcon 패키지 설치

직접 패키지를 개발하고 빌드하기 위해 필요

```
sudo apt install python3-colcon-common-extensions
```

영상으로.
(영상은 Humble 버전임에 유의!)



https://youtu.be/KR_A_qjzSmY

PinkLAB YouTube Channel

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